

- 一阶线性微分方程

$$\frac{dy}{dx} + p(x)y = Q(x)$$

$Q(x) \equiv 0$ 时 称为一阶齐次线性微分方程

齐次 $\frac{dy}{dx} = -p(x) \cdot y$

$$y=0 \text{ 或 } \frac{1}{y} dy = -p(x) dx$$

$$\Rightarrow \ln|y| = -\int_{x_0}^x p(t) dt + \tilde{C}$$

$$y = C e^{-\int_{x_0}^x p(t) dt} \quad \text{脱绝对值, 加上特解}$$

$C \in \mathbb{R}$

非齐次 $(\frac{dy}{dx} + p(x) \cdot y) e^{\int_{x_0}^x p(t) dt} = Q(x) e^{\int_{x_0}^x p(t) dt}$

$$\frac{d}{dx} (e^{\int_{x_0}^x p(t) dt} \cdot y) = Q(x) e^{\int_{x_0}^x p(t) dt}$$

$$e^{\int_{x_0}^x p(t) dt} \cdot y = \int_{x_0}^x Q(s) e^{\int_{x_0}^s p(t) dt} ds + C$$

$$\begin{aligned} \Rightarrow y &= e^{-\int_{x_0}^x p(t) dt} (C + \int_{x_0}^x Q(s) e^{\int_{x_0}^s p(t) dt} ds) \\ &= C e^{-\int_{x_0}^x p(t) dt} + e^{-\int_{x_0}^x p(t) dt} \cdot \int_{x_0}^x Q(s) e^{\int_{x_0}^s p(t) dt} ds \end{aligned}$$

① 找 $p(x), Q(x)$

② 算 $\mu(x) = e^{\int p(x) dx}$

③ 左 $\frac{d}{dx} (y \cdot \mu(x))$

④ 积分解出 y

常数变易法

设 $y = c(x) e^{-\int_{x_0}^x p(t) dt}$ 为非齐次方程解
↑
待定

$$\frac{dy}{dx} = \frac{dc(x)}{dx} \cdot e^{-\int_{x_0}^x p(t) dt} - c(x) e^{-\int_{x_0}^x p(t) dt} \cdot p(x)$$

$$\parallel$$
$$-p(x)y + Q(x) = -p(x)c(x) \cdot e^{-\int_{x_0}^x p(t) dt} + Q(x)$$

$$\frac{d}{dx} c(x) = Q(x) e^{\int_{x_0}^x p(t) dt}$$

$$c(x) = C + \int_{x_0}^x Q(s) e^{\int_{x_0}^s p(t) dt} ds$$

$$y(x) = e^{-\int_{x_0}^x p(t) dt} \cdot \left(C + \int_{x_0}^x Q(s) e^{\int_{x_0}^s p(t) dt} ds \right)$$

求 $(x+2) \frac{dy}{dx} - 2y = e^x (x+2)^3$

齐次解好求

$$\frac{dy}{dx} - \frac{2y}{x+2} = e^x (x+2)^2$$

$$\text{令 } y = c(x) (x+2)^2$$

$$\frac{dy}{dx} = c'(x) (x+2)^2 + 2c(x) (x+2) = \frac{2}{x+2} c(x) (x+2)^2 + e^x (x+2)^3$$

$$\parallel$$
$$\frac{2y}{x+2} + e^x (x+2)^3$$

$$\Rightarrow y = c(x) (x+2)^2$$

求解 $y' + y \tan x = \sec x$

$$y' = -y \tan x \Rightarrow \frac{dy}{y} = -\tan x \, dx \Rightarrow \ln|y| = \ln|\cos x| + \hat{C}$$

$$\Rightarrow y = C \cdot \cos x$$

$$\text{设 } y = C(x) \cdot \cos x \Rightarrow \frac{dy}{dx} = C'(x) \cdot \cos x - C(x) \cdot \sin x \\ = -C(x) \cdot \cos x \cdot \tan x + \sec x = \frac{1}{\cos x}$$

$$\Rightarrow C'(x) = \frac{1}{\cos^2 x} \Rightarrow C(x) = \tan x + C$$

$$\Rightarrow y = (\tan x + C) \cdot \cos x$$

解: $\frac{dy}{dx} = \frac{y}{x-y^2}$

$$\frac{dx}{dy} = \frac{1}{y} x - y$$

$$x = e^{\int \frac{1}{y} dy} (C + \int (-y) e^{-\int \frac{1}{y} dy} dy) = y(C - y)$$

$y=0$ 也是方程的解

伯努利方程

$$\frac{dy}{dx} + P(x)y = Q(x)y^\alpha$$

$$\alpha \neq 0, 1$$

$$y^{-\alpha} \frac{dy}{dx} + P(x)y^{1-\alpha} = Q(x)$$

$$\overset{||}{\frac{1}{1-\alpha} \frac{d}{dx} \frac{1}{y^{\alpha-1}} + P(x) \frac{1}{y^{\alpha-1}} = Q(x)}$$

$$\text{令 } z = \frac{1}{y^{\alpha-1}} = y^{1-\alpha} \Rightarrow \frac{1}{1-\alpha} \frac{dz}{dx} + P(x)z = Q(x)$$

一阶线性微分方程

$$\text{求 } \frac{dy}{dx} = \frac{y}{x} - xy^3$$

$$\frac{dy}{dx} - \frac{1}{x}y = -xy^3$$

$\alpha=2$ 的伯努利方程, 令 $u = y^{-1}$ 上式化为

$$\frac{du}{dx} + \frac{1}{x}u = x$$

$$u = e^{-\int \frac{1}{x} dx} (C + \int x e^{\int \frac{1}{x} dx} dx)$$

$$= \frac{1}{x} (C + \frac{1}{3}x^3) = \frac{C}{x} + \frac{x^2}{3}$$

$$\frac{1}{y} = \frac{C}{x} + \frac{x^2}{3}$$

